

Younes Essafouri

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EDUCATION

Institut de Mathématiques de Toulouse (IMT) <i>PhD in Explainable AI for Weather Forecasting</i> • Supervisors: Laurent Risser; Laure Raynaud; Luciano Drozda. • Explaining neural weather prediction models: which parts of the atmosphere a forecast relies on, how that influence propagates across autoregressive steps.	Oct 2025 – Oct 2028 (expected) Toulouse, France
KTH Royal Institute of Technology <i>Exchange Program in Machine Learning</i>	Aug 2024 – Jan 2025 Stockholm, Sweden
Grenoble INP – Ensimag, Université Grenoble Alpes <i>Engineer's Degree (M.Sc. equivalent) in Computer Science and Applied Mathematics</i>	Aug 2022 – Aug 2025 Grenoble, France
Preparatory classes (CPGE) <i>Intensive two-year program in mathematics and physics preparing for the Grandes Écoles entrance exams</i>	Sep 2020 – Jul 2022

PUBLICATIONS

Explanation of Dynamic Physical Field Predictions using WassersteinGrad: Application to Autoregressive Weather Forecasting Y. Essafouri, L. Raynaud, L. Drozda, L. Risser arXiv:2604.22580 Project page Code	<i>NeurIPS 2026</i>
A Framework for Explainable AI in Weather Forecasting: Diagnosing Deep Learning Models via Gradient-Based Attributions Y. Essafouri, C. Seznec, L. Drozda, L. Raynaud, L. Risser doi:10.5194/egusphere-egu26-4039	<i>EGU 2026, Poster</i>

RESEARCH EXPERIENCE

Research Intern <i>Dassault Systèmes</i> • Developed physics-informed neural network (PINN) surrogates for cardiac heat-transfer simulations.	Feb 2025 – Aug 2025 Vélizy-Villacoublay, France
AI Researcher <i>KTH AI Society</i> • Worked with KTH researchers on physics-informed machine learning for fluid dynamics and surrogate modeling.	Oct 2024 – Jan 2025 Stockholm, Sweden
Machine Learning Research Intern <i>Laboratoire Hubert Curien</i> • Developed PINNs to replace a computationally expensive solver for the Kuramoto–Sivashinsky equation. • Explored neural operators, focusing on Fourier Neural Operators (FNOs), for efficient spatio-temporal modeling.	Jun 2024 – Aug 2024 Saint-Étienne, France

ACADEMIC SERVICE

Journal Reviewer

- *Quarterly Journal of the Royal Meteorological Society*
- *Artificial Intelligence for the Earth Systems (AMS)*

TALKS, TEACHING

Talk: Explainable AI for Data-Driven Weather Forecasting <i>Doctoral seminar of the Stat&Opt team, Institut de Mathématiques de Toulouse</i>	2026 Toulouse, France
Teaching Assistant: Linear Models and Statistical Inference <i>Université Paul Sabatier, practical sessions in R, from least squares to confidence intervals</i>	2025 – 2026 Toulouse, France

RELEVANT COURSEWORK & SKILLS

Selected coursework: Advanced Machine Learning, Statistical Learning, Probability & Statistics, PDEs, Numerical Methods, Data Structures & Algorithms, Object-Oriented Programming.
Programming: Python, C/C++, Java, SQL, R, $\text{\LaTeX}$
ML & Scientific Computing: PyTorch, PyTorch Lightning, PyTorch Geometric, NumPy, SciPy, pandas, scikit-learn, Matplotlib
Weather & Scientific Data: Anemoi, xarray, Zarr, POT (Python Optimal Transport)
Tools & Computing: Git/GitHub, Linux/Unix, Jupyter, SLURM, HPC
Languages: English (Professional), French (Professional), Arabic (Native)